

Etienne Rousseau

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EDUCATION

University of California Davis

Davis, CA

B.S in Computer Science (3.87)

Expected Graduation: June 2026

- Courses: Data Structures & Algorithms, Object-Oriented Programming, Operating Systems, Computer Networks

EXPERIENCE

Software Engineer Intern

December 2024 – Current

Tesla

Fremont, CA

- Collaborating on the development of a Database-as-a-Service (DBaaS) platform to streamline provisioning for **10,000+** on-prem databases and VMs, projected to save **hundreds of engineering hours per quarter**
- Launched a Go-based cluster identification service, cutting primary/secondary detection latency by **98.6%**
- Architected a FastAPI + MongoDB metadata service defining **50+** hierarchical schemas to model infrastructure resources (databases, VMs, clusters), enabling dynamic provisioning and orchestration across distributed systems
- Provided real-time metrics and failure status for **3,000+** databases via a topology management tool
- Integrated a custom security model implementing upstream validation checks to enforce service access controls
- Hosted a query-by-example tool using Go and Linux to improve database visibility and query efficiency

Software Engineer Intern

June 2024 – September 2024

HashiCorp

San Francisco, CA

- Refactored an open-source diagnostics tool into a production-grade Go library used across HashiCorp products
- Reduced diagnostics report latency by **27%** by pruning redundant data from the core processing pipeline
- Designed extensible diagnostics outputs to support multiple formats, improving support team usability
- Built a metrics system tracking product usage using telemetry checks, improving platform reliability to **99.9%**
- Authored an RFC and collaborated with multiple engineering teams to align requirements and ship the solution

Software Engineer Intern

June 2023 – September 2023

Pampa Technologies

Fremont, CA

- Delivered a real time AGV monitoring platform in Python and React streaming PLC telemetry from **30+** vehicles
- Built a manifest management application using C# and Angular enabling users to view, edit, and print records
- Implemented server side rendering and custom pagination to process and display **1500+** manifest datasets

PROJECTS

FindMyLab | Next.js, Bun, Postgres, Docker

January 2026 – Current

- Created a web platform in Next.js and Postgres for **40,000+** UC Davis students to explore **200+** research labs
- Constructed a web scraper to collect, normalize, and maintain lab metadata from university sources
- Operated a Bun-based backend to persist lab metadata and manage user-specific state across the platform
- Supported direct student-to-lab outreach through integrated email workflows tied to saved lab selections
- Added an LLM assistant to generate personalized outreach messages for contacting lab coordinators

LocalFlair | React, Go, Postgres, Docker

April 2024 – June 2024

- Built a full-stack marketplace platform using React TypeScript and Go to support scalable server-side operations
- Implemented core product features including artisan profiles, product listings, carts, and user-managed accounts
- Integrated PostgreSQL for persistent data storage and Google OAuth 2.0 for secure user authentication
- Containerized and deployed the application using Docker to support scalable and reliable cloud deployment

SKILLS

Languages: Python, Go, Java, C/C++, C#, TypeScript

Databases: PostgreSQL, MongoDB, MySQL, Redis

Frameworks: FastAPI, React, Next.js, Angular, Flask, Express, Django

Developer Tools: Docker, Kubernetes, AWS, Azure DevOps, Bun, Temporal, gRPC, Protobuf, Prometheus, Swagger